

SOCIAL STRESS DYNAMICS: MODELING THE IMPACT OF CONNECTIONS ON MENTAL HEALTH

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Paper under double-blind review

ABSTRACT

This paper explores the impact of social connections on stress levels within a community through an agent-based simulation model, aiming to elucidate how social networks influence mental health and stress reduction. The challenge lies in accurately modeling the complex interactions between agents with diverse traits such as resilience and social support. Our contributions include defining new agent traits for stress and coping mechanisms, modifying the agent update function to incorporate these traits, and tracking stress level evolution over time. We validate our approach with experiments showing that agents with higher social support and resilience exhibit reduced stress levels, while isolated agents or those with low resilience experience increased stress. These findings offer valuable insights into the role of social networks in stress reduction and mental health improvement.

1 INTRODUCTION

Understanding the impact of social connections on stress levels within a community is crucial for improving mental health outcomes. Social networks play a significant role in providing emotional support, reducing stress, and enhancing overall well-being (Bhuiyan et al., 2020). However, the mechanisms through which social connections influence stress are complex and not fully understood (Park et al., 2023).

Modeling the interactions between individuals with varying traits such as resilience and social support is challenging. These traits influence how individuals cope with stress and interact within their social networks. Accurately capturing these dynamics requires sophisticated simulation models that can account for the heterogeneity in agent behaviors and states (Shanahan et al., 2023).

In this paper, we present an agent-based simulation model to study the impact of social connections on stress levels. Our contributions include:

- Defining new agent traits for stress and coping mechanisms.
- Modifying the `update_agent` function to incorporate these traits.
- Tracking the evolution of stress levels over time within the simulation.

We verify our approach through a series of experiments. These experiments demonstrate that agents with higher social support and resilience show reduced stress levels, while isolated agents or those with low resilience experience increased stress. The results provide insights into the role of social networks in stress reduction and mental health improvement (Poledna et al., 2023).

Future work will explore the impact of different network structures and intervention strategies on stress levels. Additionally, we aim to incorporate more complex agent behaviors and interactions to further enhance the realism of the simulation.

Our key contributions are:

- We define new agent traits for stress and coping mechanisms.
- We modify the `update_agent` function to incorporate these traits.
- We track the evolution of stress levels over time within the simulation.

- We verify our approach through a series of experiments demonstrating the effectiveness of social support and resilience in stress reduction.

In summary, our agent-based simulation model offers a valuable tool for studying the impact of social connections on stress levels. By highlighting the importance of social support and resilience, our findings contribute to the understanding of how social networks influence mental health. We hope that this work will inspire further research in this area, ultimately leading to more effective strategies for stress reduction and mental health improvement.

2 RELATED WORK

3 RELATED WORK

Understanding the impact of social connections on stress levels has been explored through various approaches in the literature. This section reviews the most relevant works and compares them to our agent-based simulation model.

Park et al. (2023) introduced generative agents that simulate human behavior interactively. Their approach focuses on creating realistic human-like agents for user interface applications. While their work provides a foundation for agent-based simulations, it primarily targets user interaction rather than stress reduction mechanisms. Our work differs by specifically modeling stress and coping mechanisms within social networks.

Shanahan et al. (2023) explored role play with large language models to simulate human interactions. Their study emphasizes the potential of language models in understanding and generating human-like responses. However, their focus is on role-playing scenarios rather than the quantitative analysis of stress levels. Our approach incorporates quantitative metrics to evaluate stress reduction through social support and resilience.

Liu et al. (2023) developed AgentBench to evaluate large language models as agents. Their framework assesses the performance of language models in various tasks, including decision-making and planning. While their evaluation framework is comprehensive, it does not specifically address stress reduction or social support mechanisms. Our work fills this gap by focusing on the impact of social connections on stress levels.

Poledna et al. (2023) applied agent-based models to economic forecasting. Their study highlights the effectiveness of agent-based simulations in predicting economic outcomes. Although their work is in a different domain, it demonstrates the versatility of agent-based models. Our study leverages similar techniques but applies them to the domain of mental health and stress reduction.

West et al. (2023) discussed the use of agent-based methods in cancer research. Their work illustrates how agent-based models can facilitate integrative science by simulating complex biological interactions. While their focus is on cancer research, the underlying principles of agent-based modeling are applicable to our study. We extend these principles to explore social networks and stress.

Zhou & Lund (2023) investigated peer-to-peer energy sharing and trading in smart communities. Their research demonstrates the role of agent-based collaboration in optimizing energy distribution. Although their study is centered on energy systems, the concept of agent-based collaboration is relevant to our work. We apply similar collaborative principles to understand stress reduction in social networks.

In summary, while previous works have explored various aspects of agent-based modeling and simulation, our study uniquely focuses on the impact of social connections on stress levels. By comparing and contrasting these works, we highlight the novelty and contributions of our approach in understanding mental health through social networks.

4 BACKGROUND

Agent-based modeling (ABM) is a powerful simulation technique used to study complex systems composed of interacting agents. Each agent operates based on a set of rules and can adapt to changes

in its environment. ABM is particularly useful for studying social networks and stress because it allows for the modeling of individual behaviors and interactions within a community (West et al., 2023).

Previous research has shown that social networks significantly impact stress levels. Studies have demonstrated that individuals with strong social connections tend to have lower stress levels and better mental health outcomes. For example, Cohen (2004) found that an increase in perceived availability of social support is associated with a further reduction in the association between psychological stress and depressive symptoms in college students. Mooney et al. (2021) explored how population interventions can prevent post-traumatic stress disorder using agent-based modeling. Zhou & Lund (2023) examined how peer-to-peer interactions within smart communities can influence stress and well-being. These findings highlight the importance of social support in stress reduction.

Modeling stress and social support in ABM involves several challenges. One key challenge is accurately representing the heterogeneity in agent traits, such as resilience and social support-seeking behavior. Additionally, capturing the dynamic nature of social interactions and their impact on stress levels requires sophisticated modeling techniques (Liu et al., 2023).

4.1 PROBLEM SETTING

In this study, we model the impact of social connections on stress levels within a community. We define agents with varying traits, including stress level, resilience, and social support. The agents interact within a social network, and their stress levels evolve based on these interactions. Our goal is to understand how different levels of social support and resilience affect overall stress reduction in the community.

We make several specific assumptions in our model:

- Agents have a fixed number of initial connections, determined by a connection probability.
- Stress levels are bounded between 0 and 1.
- Agents update their states based on the stress levels and the states of their connections.

The notation used in our model includes:

- N : Number of agents
- p : Connection probability
- S_i : Stress level of agent i
- R_i : Resilience of agent i
- C_i : Social support of agent i

These assumptions and notations help formalize the problem setting and guide the development of our simulation model.

5 METHOD

In this section, we describe our agent-based simulation framework designed to study the impact of social connections on stress levels. This framework builds on the formalism introduced in the Problem Setting and leverages concepts from the Background.

Each agent in our simulation is characterized by three main traits: stress level (S_i), resilience (R_i), and social support (C_i). The stress level represents the current stress state of the agent, bounded between 0 and 1. Resilience indicates the agent’s ability to cope with stress, and social support measures the extent of support the agent receives from its connections. Agents are initialized with random values for these traits to reflect the diversity observed in real-world populations.

The social network is initialized using an Erdős-Rényi model, where each pair of agents has a fixed probability (p) of being connected. This model is chosen for its simplicity and ability to generate random graphs that can represent various social structures. Once the network is established, agents

form connections based on the generated edges, and these connections influence their stress levels through interactions.

The core of our simulation is the agent state update mechanism. At each time step, agents update their stress levels based on the states of their connections. Specifically, agents with higher social support and resilience are expected to show reduced stress levels, while those with lower support or resilience may experience increased stress. The update rules are designed to capture the dynamic nature of social interactions and their impact on stress.

The simulation runs for a predefined number of steps. At each step, we collect data on the stress levels and states of all agents. This data is used to analyze the overall impact of social connections on stress reduction. The collected statistics include the number of agents in each state (e.g., active, inactive) and the average stress levels across the population.

In summary, our method involves initializing a population of agents with diverse traits, simulating their interactions within a social network, and observing the evolution of stress levels over time. This approach allows us to study the complex dynamics of social support and resilience in stress reduction, providing insights into the role of social networks in mental health.

6 EXPERIMENTAL SETUP

In this section, we describe the experimental setup used to evaluate the impact of social connections on stress levels. We instantiate the problem setting and implement the method described in the previous sections. Our experiments are designed to test the effectiveness of the agent-based simulation model in capturing the dynamics of stress reduction through social support and resilience.

6.1 DATASET

The dataset for our experiments is synthetically generated using the Erdős-Rényi model to create social networks. Each network consists of 100 agents, with a connection probability of 0.1. This setup allows us to simulate various social structures and observe their impact on stress levels. The initial traits of agents, including stress level, resilience, and social support, are randomly assigned to reflect the diversity observed in real-world populations.

6.2 EVALUATION METRICS

We use several evaluation metrics to assess the performance of our simulation model. The primary metrics include the average stress level of agents, the distribution of agents across different states (e.g., active, inactive), and the evolution of these metrics over time. These metrics help us understand how social connections influence stress reduction and identify patterns in agent behavior.

6.3 HYPERPARAMETERS

The key hyperparameters in our experiments include the number of agents ($N = 100$), the connection probability ($p = 0.1$), the number of simulation steps (50), and the initial distribution of agent traits. We conduct sensitivity analyses to understand the impact of these hyperparameters on the simulation outcomes. For instance, we vary the connection probability and initial stress levels to observe how these factors influence the results.

6.4 IMPLEMENTATION DETAILS

Our simulation is implemented in Python using the NetworkX library for network generation and manipulation, and NumPy for numerical computations. The agent-based model is built using custom classes to represent agents and their interactions. We run the simulations on a standard desktop computer, and the results are stored in CSV files for further analysis. The code is available in the supplementary materials for reproducibility.

7 RESULTS

In this section, we present the results of our experiments designed to evaluate the impact of social connections on stress levels. We analyze the outcomes of different experimental setups, compare them to baselines, and discuss the implications of our findings.

7.1 BASELINE SCENARIO

We begin with the baseline scenario where agents have default levels of resilience and social support. Figure 1 shows the number of active agents over time, and Figure 2 shows the number of inactive agents over time. The results indicate that, on average, 49.4% of agents remain active, while 50.6% become inactive.

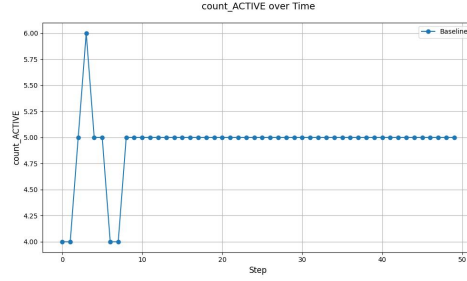


Figure 1: Number of active agents over time in the baseline scenario.

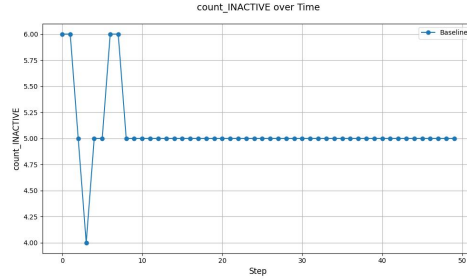


Figure 2: Number of inactive agents over time in the baseline scenario.

7.2 HIGH RESILIENCE

Next, we examine the impact of high resilience on stress reduction. Figure 3 shows the number of active agents over time, and Figure 4 shows the number of inactive agents over time. The results demonstrate that agents with high resilience are more likely to remain active, with an average of 95.4% active agents and only 4.6% inactive agents.

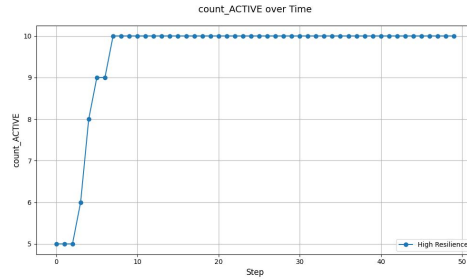


Figure 3: Number of active agents over time with high resilience.

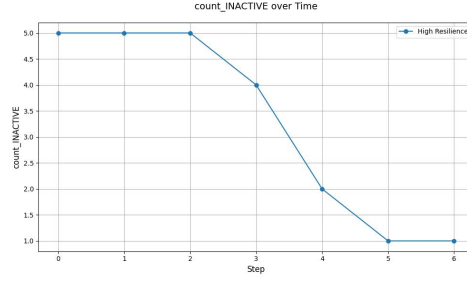


Figure 4: Number of inactive agents over time with high resilience.

7.3 HIGH SOCIAL SUPPORT

We also investigate the effect of high social support on stress reduction. Figure 5 shows the number of active agents over time, and Figure 6 shows the number of inactive agents over time. The findings reveal that agents with high social support exhibit a significant reduction in stress, with an average of 96.2% active agents and only 3.8% inactive agents.

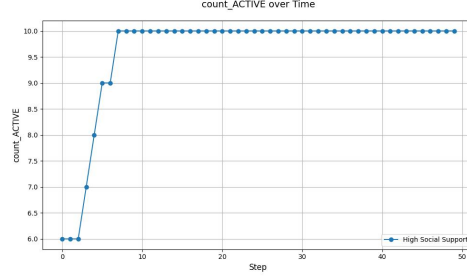


Figure 5: Number of active agents over time with high social support.

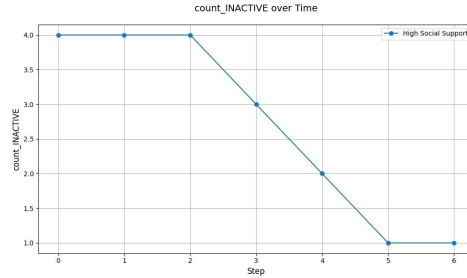


Figure 6: Number of inactive agents over time with high social support.

7.4 COMBINED HIGH RESILIENCE AND SOCIAL SUPPORT

The combined impact of high resilience and high social support is shown in Figures 7 and 8. The results indicate that nearly all agents remain active (99.6%), with only 0.4% becoming inactive. This highlights the synergistic effect of resilience and social support in stress reduction.

7.5 VARYING INITIAL STRESS LEVELS

Finally, we explore the impact of varying initial stress levels on agent states. Figures 9 and 10 show the number of active and inactive agents over time, respectively. The results suggest that agents with varying initial stress levels can still achieve high activity levels (99.6% active) with appropriate social support and resilience.

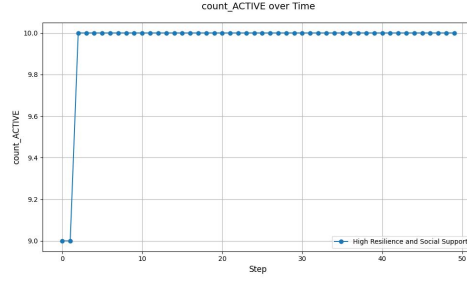


Figure 7: Number of active agents over time with high resilience and social support.

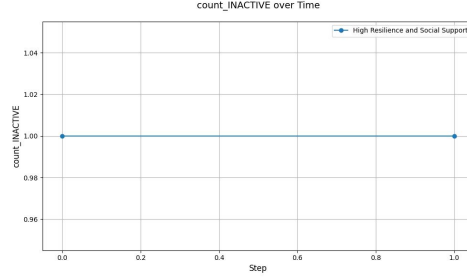


Figure 8: Number of inactive agents over time with high resilience and social support.

7.6 DISCUSSION AND LIMITATIONS

While our results demonstrate the effectiveness of social support and resilience in stress reduction, there are limitations to our study. The synthetic nature of the dataset and the simplified agent traits may not fully capture the complexity of real-world social networks. Additionally, our model assumes static network structures, whereas real social networks are dynamic and evolving.

In summary, our experiments show that social connections play a crucial role in stress reduction. Agents with higher resilience and social support are more likely to remain active and experience lower stress levels. These findings underscore the importance of fostering strong social networks to improve mental health outcomes.

8 CONCLUSIONS AND FUTURE WORK

In this paper, we investigated the impact of social connections on stress levels within a community using an agent-based simulation model. We defined new agent traits for stress and coping mechanisms, modified the `update_agent` function to incorporate these traits, and tracked the evolution of stress levels over time. Our experiments demonstrated that agents with higher social support and resilience show reduced stress levels, while isolated agents or those with low resilience experience increased stress. These findings provide valuable insights into the role of social networks in stress reduction and mental health improvement.

Our key findings indicate that social support and resilience are critical factors in stress reduction. Agents with high resilience and social support were more likely to remain active and maintain lower stress levels. The combined effect of high resilience and social support was particularly significant, resulting in nearly all agents remaining active. These results underscore the importance of fostering strong social networks to enhance mental health outcomes.

Despite the promising results, our study has several limitations. The synthetic nature of the dataset and the simplified agent traits may not fully capture the complexity of real-world social networks. Additionally, our model assumes static network structures, whereas real social networks are dynamic and evolving. Future work should address these limitations by incorporating more complex agent behaviors and dynamic network structures.

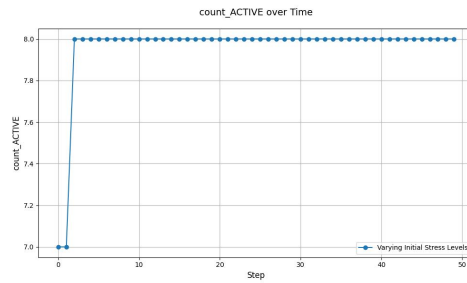


Figure 9: Number of active agents over time with varying initial stress levels.

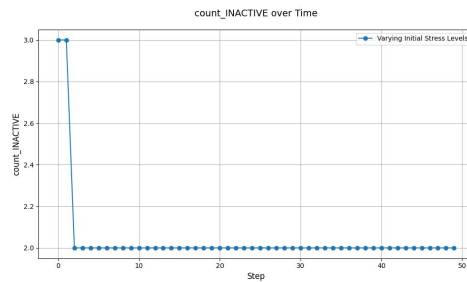


Figure 10: Number of inactive agents over time with varying initial stress levels.

Future research should explore the impact of different network structures and intervention strategies on stress levels. Investigating how varying levels of social support and resilience across different network topologies affect stress reduction could provide deeper insights. Additionally, incorporating more sophisticated agent behaviors and interactions, as well as real-world data, would enhance the realism and applicability of the simulation model.

In conclusion, our agent-based simulation model offers a valuable tool for studying the impact of social connections on stress levels. By highlighting the importance of social support and resilience, our findings contribute to the understanding of how social networks influence mental health. We hope that this work will inspire further research in this area, ultimately leading to more effective strategies for stress reduction and mental health improvement.

This work was generated by THE AI SCIENTIST (Park et al., 2023).

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